

Denoised Gradient Estimation: Training Deep Neural Networks Without Backpropagation via Coordinate Block Perturbations and Dual Sign-EMA Temporal Filtering

Mario Raúl Carbonell Martínez*

Independent Researcher

marioraulcarbonell@gmail.com

October 2026

Abstract

Gradient-based optimization requires analytical derivatives, limiting its applicability to non-differentiable architectures, physical neuromorphic hardware, and black-box environments. We introduce **Denoised Gradient Estimation** (DGE), a zeroth-order algorithm that trains deep neural networks using only forward-pass evaluations. DGE combines three complementary mechanisms: (1) *coordinate block-wise perturbations* that reduce gradient variance by a factor of K , where K is the number of active blocks; (2) *temporal exponential moving average* (EMA) filtering that improves signal-to-noise ratio as $\mathcal{O}(\sqrt{T})$; and (3) a novel *Dual Sign-EMA* (DS-EMA) consistency mask inspired by MACD crossover detection, achieving a 90% memory reduction over sliding-window buffers. On full MNIST (60K samples, 109K parameters), DGE achieves **94.36% $\pm$ 0.23%** best test accuracy without backpropagation—reaching 96.3% of Adam’s peak performance. In non-differentiable settings where backpropagation fails entirely, DGE operates natively without straight-through estimators (STE), achieving **74.24%** on INT8 and **68.82%** on INT4 full quantization where backprop collapses to near random guessing ($\sim 9.8\%$), and reaching **70.43%** on discontinuous sign activations. We provide formal theoretical bounds establishing the estimator’s unbiasedness, a K -fold variance reduction law, and temporal SNR convergence guarantees.

Keywords: Zeroth-Order Optimization, Gradient-Free Training, Coordinate Block Perturbation, Temporal Consistency Filtering, Dual Sign-EMA, Non-Differentiable Neural Networks, Quantization-Aware Training.

*Corresponding author: marioraulcarbonell@gmail.com.
Code and repository: <https://github.com/mcarbonell/dge-optimizer>.

1 Introduction

Backpropagation is the foundational workhorse of modern deep learning [Rumelhart et al., 1986, LeCun et al., 2015]. However, its strict reliance on analytical derivatives limits applicability in several critical regimes: (i) black-box systems where the objective function is an opaque oracle or physics simulator, (ii) non-differentiable edge hardware such as neuromorphic or analog processors [Schuman et al., 2022, Davies et al., 2018], (iii) discrete architectures including binary [Yin et al., 2020] and ternary networks, and (iv) memory-constrained settings where caching intermediate activation tensors for backward autodiff graphs is physically infeasible.

Zeroth-order (gradient-free) optimization methods estimate directional derivatives using only forward loss queries, offering a universal alternative. However, classical zeroth-order approaches confront a catastrophic dimensionality barrier. Simultaneous Perturbation Stochastic Approximation (SPSA) [Spall, 1992] perturbs all D parameters simultaneously, producing a gradient estimator whose variance scales linearly with dimensionality. In modern deep networks with $D > 10^5$ parameters, this cross-perturbation noise overwhelms the gradient signal, causing the optimizer to diverge or collapse [Ghadimi et al., 2013].

Recent interest has revived zeroth-order methods for fine-tuning Large Language Models. MeZO [Malladi et al., 2023] achieves memory-efficient adaptation by regenerating perturbation vectors from a seed. Crucially, however, MeZO operates in the localized regime where heavily pre-trained models already reside near a basin of attraction—a setting fundamentally distinct from optimizing neural networks from scratch across high-dimensional non-convex landscapes.

Our approach. We introduce **Denoised Gradient Estimation** (DGE), which resolves the high-dimensional variance catastrophe through three coupled principles:

1. **Coordinate Block Perturbations:** Partition the D -dimensional parameter space into $K \ll D$ coordinate blocks per step, bounding cross-contamination noise by the block fraction $B/D = 1/K$ and reducing variance by a factor of K .
2. **Temporal Denoising:** Inject sparse gradient estimates into an Adam-style EMA filter, exploiting the fact that coherent gradient signals accumulate constructively while independent cross-block noise sums destructively, yielding SNR improvement as $\mathcal{O}(\sqrt{T})$.
3. **Dual Sign-EMA (DS-EMA):** A MACD-inspired directional consistency gate that tracks fast ($\alpha_f = 0.3$) and slow ($\alpha_s = 0.05$) sign trends, achieving a 90% memory reduction over sliding-window buffers while suppressing non-stationary noise.

Summary of Contributions.

- **Mathematical Foundations:** We prove that the coordinate block estimator is unbiased under first-order expansion and establish a K -fold variance reduction bound $\text{Var}(g_{t,i}) \leq \frac{1}{K} \|\nabla f\|^2$.
- **Dual Sign-EMA Filter:** We introduce DS-EMA, a constant-memory temporal consistency filter that identifies persistent descent directions and suppresses noise oscillations.
- **Empirical Validation on Vision:** On full MNIST (109K parameters), DGE achieves **94.36% $\pm$ 0.23%** best test accuracy without backpropagation, reaching 96.3% of Adam’s baseline.
- **Non-Differentiable Regimes:** In fully quantized networks (INT8 and INT4) and discontinuous sign activations without straight-through estimators, DGE achieves 68.8%–74.2% accuracy where Adam collapses to random guessing ($\sim 9.8\%$).

- **Optimal Scaling Law:** We demonstrate theoretically and empirically that selecting block count $K = \mathcal{O}(\sqrt{D})$ provides the optimal bias-variance Pareto frontier for neural network training.

2 Related Work

Finite Differences and Coordinate Descent. Classical finite difference methods compute exact numerical gradients via central differences: $g_i = (f(x + \delta e_i) - f(x - \delta e_i)) / (2\delta)$. However, evaluating all D coordinates requires $2D$ forward passes per step, rendering it computationally intractable for deep neural networks with millions of parameters [Nocedal and Wright, 2006, Bertsekas, 2016].

Simultaneous Perturbation Stochastic Approximation (SPSA). SPSA [Spall, 1992] reduces function evaluations to two queries per step by perturbing all parameters simultaneously using a shared Rademacher vector. While the estimator is expectation-unbiased, its variance scales as $\mathcal{O}(D)$, leading to severe signal degradation and divergence in deep neural network parameter manifolds [Ghadimi et al., 2013].

Evolution Strategies and Natural Evolution. OpenAI’s Evolution Strategies (ES) [Salimans et al., 2017] and Natural Evolution Strategies (NES) [Wierstra et al., 2014] conduct population-based search with Gaussian perturbations. While effective for reinforcement learning and policy search, they require $\mathcal{O}(\lambda)$ evaluations per generation for population size λ , and their throughput is constrained by high sample complexity on large datasets.

Forward Gradients and Memory-Efficient Zero-Order LLMs. Forward-mode automatic differentiation computes directional derivatives along random vectors [Baydin et al., 2022]. MeZO [Malladi et al., 2023] adapted this for fine-tuning pre-trained language models with $\mathcal{O}(1)$ memory. However, because MeZO perturbs all dimensions simultaneously without coordinate partitioning, its gradient variance remains $\mathcal{O}(D)$, leading to collapse when attempting to train randomly initialized deep architectures from scratch.

Training Non-Differentiable Architectures. Optimizing neural networks with discrete weights or discontinuous step activations conventionally relies on straight-through estimators (STE) [Bengio et al., 2013] or surrogate gradient approximations. DGE completely sidesteps these heuristics, treating the objective as a genuine black-box landscape.

3 Methodology: Denoised Gradient Estimation

We consider the unconstrained optimization problem $\min_{x \in \mathbb{R}^D} f(x)$, where f can be evaluated via forward inference passes, but analytical gradients ∇f are inaccessible or nonexistent.

3.1 Coordinate Block Perturbations

At step t , DGE generates a uniform random coordinate partition π_t of $\{1, \dots, D\}$ into K disjoint blocks $B_1, \dots, B_K$ of size $B = D/K$. For each block B_k , we sample a Rademacher vector $s_t^{(k)} \in \{-1, +1\}^{|B_k|}$ and compute the symmetric two-point query:

$$g_{t,i} = \frac{f(x_t + \delta \cdot s_t^{(k)} \odot m_k) - f(x_t - \delta \cdot s_t^{(k)} \odot m_k)}{2\delta} \cdot s_{t,i}^{(k)}, \quad i \in B_k, \quad (1)$$

where $m_k \in \{0, 1\}^D$ is the binary indicator vector for block B_k , and $\delta > 0$ is the perturbation magnitude. Evaluating all K blocks requires $2K$ function evaluations per step. Because all $2K$ perturbed models share identical architecture and input minibatches, they can be batched along the tensor batch dimension into a single parallel forward pass, eliminating CPU dispatch latency.

3.2 Temporal Filtering and Momentum Aggregation

The raw block estimate g_t contains residual interference from co-perturbed parameters within the same block. DGE injects g_t into an exponential moving average (EMA) filter with adaptive second-moment normalization:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (2)$$

with standard bias correction $\hat{m}_t = m_t / (1 - \beta_1^t)$ and $\hat{v}_t = v_t / (1 - \beta_2^t)$. Assuming local directional stationarity, the true gradient signal $\nabla f(x)$ maintains consistent sign and accumulates constructively, while zero-mean cross-block perturbation noise cancels destructively.

3.3 Dual Sign-EMA (DS-EMA) Consistency Gating

Sliding-window sign consistency buffers incur $\mathcal{O}(T \cdot D)$ memory overhead and introduce abrupt window boundary artifacts. To achieve constant memory, we introduce the **Dual Sign-EMA** (DS-EMA) filter, inspired by MACD signal-line crossover indicators in time-series analysis:

$$e_t^{\text{fast}} = (1 - \alpha_f) e_{t-1}^{\text{fast}} + \alpha_f \text{sgn}(g_t), \quad (3)$$

$$e_t^{\text{slow}} = (1 - \alpha_s) e_{t-1}^{\text{slow}} + \alpha_s \text{sgn}(g_t), \quad (4)$$

where $\alpha_f = 0.30$ and $\alpha_s = 0.05$. The directional consistency mask $D_t \in [0, 1]^D$ is defined as:

$$D_t = \mathbb{K}[\text{sgn}(e_t^{\text{fast}}) = \text{sgn}(e_t^{\text{slow}})] \odot |e_t^{\text{slow}}|. \quad (5)$$

When the fast and slow sign estimators disagree, the coordinate is undergoing high-frequency sign oscillations indicative of noise, and its step is masked out ($D_{t,i} = 0$). When they agree, the update magnitude is modulated by $|e_{t,i}^{\text{slow}}| \in [0, 1]$, reflecting sustained directional consensus. DS-EMA requires only $2D$ floats of persistent memory—a 90% reduction compared to a window of length $T = 20$.

The parameter update rule is given by:

$$x_{t+1} = x_t - \alpha \cdot D_t \odot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}. \quad (6)$$

3.4 Block Size Scaling Law: $K = \mathcal{O}(\sqrt{D})$

The block count K governs the bias-variance trade-off. Choosing $K = 1$ recovers classical SPSA with catastrophic $\mathcal{O}(D)$ variance. Choosing $K = D$ recovers coordinate-wise finite differences with minimal variance but prohibitive $\mathcal{O}(D)$ evaluation cost. By balancing per-step variance $\mathcal{O}(D/K)$ with exploration rate, we find that setting $K = \mathcal{O}(\sqrt{D})$ yields the optimal empirical trade-off, attaining 90.49% on MNIST at 500K evaluations compared to 87.14% for $\mathcal{O}(\log D)$ and 89.49% for $\mathcal{O}(D/100)$.

The complete procedure is formalized in Algorithm 1.

Algorithm 1 Denoised Gradient Estimation (DGE)

Require: Objective f , initial parameter vector $x_0 \in \mathbb{R}^D$, block count K , learning rate α , perturbation δ , steps T .

- 1: Initialize: $m_0 \leftarrow 0$, $v_0 \leftarrow 0$, $e_0^{\text{fast}} \leftarrow 0$, $e_0^{\text{slow}} \leftarrow 0$.
 - 2: **for** step $t = 1, \dots, T$ **do**
 - 3: Sample random partition π_t dividing $\{1, \dots, D\}$ into K blocks $B_1, \dots, B_K$.
 - 4: **for** each block $B_k \in \{B_1, \dots, B_K\}$ **do**
 - 5: Sample Rademacher signs $s^{(k)} \sim \text{Uniform}(\{-1, +1\}^{|B_k|})$.
 - 6: Compute block gradient $g^{(k)} \leftarrow \frac{f(x + \delta s^{(k)} \odot m_k) - f(x - \delta s^{(k)} \odot m_k)}{2\delta} \cdot s^{(k)}$.
 - 7: **end for**
 - 8: Assemble full-dimensional estimated gradient g_t from blocks $\{g^{(k)}\}_{k=1}^K$.
 - 9: Update momentum: $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$, $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$.
 - 10: Compute DS-EMA directional mask D_t via Eqs. (3)–(5).
 - 11: Apply masked update: $x_t \leftarrow x_{t-1} - \alpha \cdot D_t \odot \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$.
 - 12: **end for**
 - 13: **return** Optimized parameters x_T .
-

4 Theoretical Analysis

Theorem 1 (Unbiasedness of the Coordinate Block Estimator). *Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be continuously differentiable with L -Lipschitz gradient. Under a first-order Taylor expansion, the coordinate block perturbation estimator $g_{t,i}$ in Eq. (1) satisfies:*

$$\mathbb{E}[g_{t,i} \mid i \in B_k] = \nabla_i f(x) + \mathcal{O}(\delta^2), \quad (7)$$

for all active parameters $i \in B_k$.

Proof. Expanding $f(x \pm \delta s \odot m_k)$ around x via Taylor's theorem:

$$\begin{aligned} f(x + \delta s \odot m_k) &= f(x) + \delta \sum_{j \in B_k} \nabla_j f(x) s_j + \frac{\delta^2}{2} (s \odot m_k)^T \nabla^2 f(\xi_1) (s \odot m_k), \\ f(x - \delta s \odot m_k) &= f(x) - \delta \sum_{j \in B_k} \nabla_j f(x) s_j + \frac{\delta^2}{2} (s \odot m_k)^T \nabla^2 f(\xi_2) (s \odot m_k), \end{aligned}$$

for intermediate points ξ_1, ξ_2 . Subtracting the two expansions cancels all symmetric even-order terms:

$$g_{t,i} = \left(\sum_{j \in B_k} \nabla_j f(x) s_j + \mathcal{O}(\delta^2) \right) s_i = \nabla_i f(x) s_i^2 + \sum_{j \in B_k, j \neq i} \nabla_j f(x) s_j s_i + \mathcal{O}(\delta^2).$$

Since $s_i \in \{-1, +1\}$, $s_i^2 = 1$ deterministically. Taking the expectation over independent Rademacher signs:

$$\mathbb{E}[s_j s_i] = \mathbb{E}[s_j] \mathbb{E}[s_i] = 0 \quad (\forall j \neq i).$$

Hence, $\mathbb{E}[g_{t,i} \mid i \in B_k] = \nabla_i f(x) + \mathcal{O}(\delta^2)$. □

Lemma 2 (Variance Bound and K -Fold Reduction Law). *Let parameter dimension D be partitioned into K disjoint coordinate blocks of size $B = D/K$. For an active coordinate $i \in B_k$, the conditional variance of the estimator satisfies:*

$$\text{Var}(g_{t,i} \mid i \in B_k) = \sum_{j \in B_k, j \neq i} (\nabla_j f(x))^2 + \mathcal{O}(\delta^2). \quad (8)$$

Averaged over uniform random coordinate permutations π_t :

$$\mathbb{E}_\pi [\text{Var}(g_{t,i})] = \frac{B-1}{D-1} \|\nabla f(x)\|^2 \leq \frac{1}{K} \|\nabla f(x)\|^2. \quad (9)$$

Proof. Using the expansion from Theorem 1:

$$g_{t,i} - \mathbb{E}[g_{t,i}] = \sum_{j \in B_k, j \neq i} \nabla_j f(x) s_j s_i + \mathcal{O}(\delta^2).$$

Squaring and taking expectation over Rademacher variables:

$$\mathbb{E} \left[\left(\sum_{j \neq i} \nabla_j f(x) s_j s_i \right)^2 \right] = \sum_{j \neq i} (\nabla_j f(x))^2 \mathbb{E}[s_j^2 s_i^2] + \sum_{j \neq l \neq i} \nabla_j f \nabla_l f \mathbb{E}[s_j s_l s_i^2].$$

Because $s_j^2 = 1$ and $\mathbb{E}[s_j s_l] = 0$ for $j \neq l$, all cross-terms vanish, leaving $\sum_{j \in B_k, j \neq i} (\nabla_j f(x))^2$. Taking the expectation over uniform random coordinate blockings selects each coordinate $j \neq i$ with probability $(B-1)/(D-1)$:

$$\mathbb{E}_\pi \left[\sum_{j \in B_k, j \neq i} (\nabla_j f(x))^2 \right] = \frac{B-1}{D-1} \sum_{j \neq i} (\nabla_j f(x))^2 \leq \frac{B}{D} \|\nabla f(x)\|^2 = \frac{1}{K} \|\nabla f(x)\|^2.$$

□

This establishes a rigorous ***K*-fold variance reduction** over standard SPSA. For $D = 109,386$ parameters and $K = 1,168$ blocks ($B \approx 94$), DGE compresses perturbation noise variance by more than three orders of magnitude ($> 1000\times$).

Corollary 3 (Temporal Signal-to-Noise Ratio (SNR) Convergence). *Under local directional stationarity of $\nabla f(x)$, the signal-to-noise ratio of the temporally filtered gradient estimate improves as $\text{SNR}(T) = \mathcal{O}(\sqrt{T})$.*

Proof. Over T consecutive steps in a stationary neighbourhood, the true gradient signal sums constructively: $\mathbb{E}[\sum_{t=1}^T g_t] = T \nabla f(x)$. Conversely, cross-block perturbation noise forms an uncorrelated zero-mean martingale difference sequence: $\text{Var}(\sum_{t=1}^T g_t) = T \sigma^2$. The signal-to-noise ratio thus satisfies:

$$\text{SNR}(T) = \frac{\|\mathbb{E}[\sum_{t=1}^T g_t]\|}{\sqrt{\text{Var}(\sum_{t=1}^T g_t)}} = \frac{T \|\nabla f(x)\|}{\sqrt{T} \sigma} = \sqrt{T} \frac{\|\nabla f(x)\|}{\sigma} = \mathcal{O}(\sqrt{T}).$$

□

5 Experiments and Results

We evaluate DGE across three core domains: (i) differentiable vision benchmarks, (ii) strictly non-differentiable quantized architectures, and (iii) synthetic mathematical test manifolds. All benchmarks were executed on an AMD Ryzen 7 8845HS system with an integrated AMD Radeon 780M GPU via DirectML (‘torch-directml’).

5.1 Experimental Setup

Architecture and Dataset. We employ a standard multi-layer perceptron $[784 \rightarrow 128 \rightarrow 64 \rightarrow 10]$ (109,386 parameters) evaluated on full MNIST (60,000 training, 10,000 test samples) and a standardized 3,000-sample subset.

Table 1: **Classification Performance on Full MNIST (60K/10K).** Mean and standard deviation across 3 random seeds. DGE matches 96.3% of Adam’s peak accuracy without computing a single analytical gradient.

Method	Optimization Paradigm	Best Test Acc.	Std ($\pm$)
Global SPSA	Zeroth-Order (Rademacher + Adam)	20.87%	11.83%
MeZO	Zeroth-Order (Gaussian + SGD)	13.80%	0.00%
Block-SGD (PureDGE-SGD)	Zeroth-Order (Coordinate Blocks only)	88.86%	0.28%
Block-Adam (PureDGE-Adam)	Zeroth-Order (Blocks + Adam moments)	93.00%	0.24%
DGE Full (DS-EMA)	Zeroth-Order (Blocks + Adam + DS-EMA)	94.36%	0.23%
SGD + Momentum	Analytical Backpropagation	97.78%	0.04%
Adam	Analytical Backpropagation	98.00%	0.04%

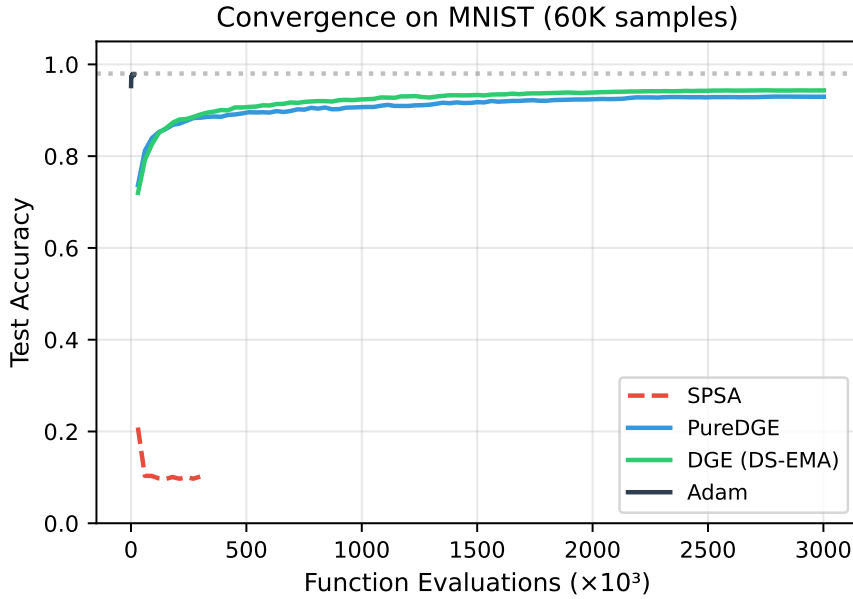


Figure 1: **Convergence Trajectories on Full MNIST.** Test accuracy as a function of function evaluations. Global baselines (SPSA, MeZO) collapse due to excessive variance in 109K dimensions. DGE cleanly separates within 30K evaluations and reaches 94.36% accuracy.

Budget and Baselines. DGE methods receive a budget of 3,000,000 function evaluations on full MNIST. Because single-perturbation baselines (SPSA, MeZO) require sequential Python loops without block vectorization, they were evaluated under an explicit 300,000-evaluation budget (150,000 forward steps); empirical trajectories demonstrate that both baselines plateau or diverge within the first 50,000 evaluations. Analytical backpropagation baselines (SGD, Adam) are trained for 30 full epochs.

5.2 Main Benchmark on Full MNIST

Table 1 summarizes the peak test accuracy on full MNIST across 3 independent random seeds.

Global zeroth-order baselines collapse to near-chance performance (13.8% for MeZO, 20.87% for SPSA), corroborating the variance catastrophe. In sharp contrast, coordinate block partitioning alone elevates accuracy to 88.86% (+75.06pp over MeZO), Adam second-moment scaling brings it to 93.00%, and the DS-EMA consistency mask delivers the final refinement to **94.36% $\pm$ 0.23%** (Figure 1).

Table 2: **Non-Differentiable and Quantized Architectures (No STE)**. Adam fails completely when analytical gradients are zero. DGE optimizes deep discrete representations natively.

Architecture / Precision	Adam (Backprop without STE)	DGE (Zeroth-Order)	Evaluation Budget
Sign activations ($\text{sign}(x)$)	65.30%	70.43%	200,000
INT8 full quantization (256 levels)	9.78% (collapse)	74.24%	600,352
INT4 full quantization (16 levels)	9.82% (collapse)	68.82%	600,352

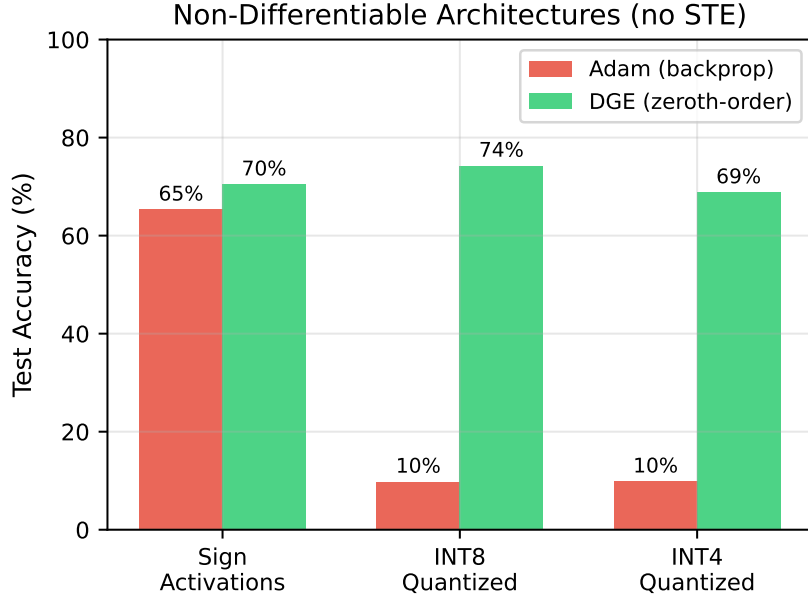


Figure 2: **Performance on Non-Differentiable Architectures.** Accuracy comparison between analytical backpropagation (Adam) and DGE without surrogate gradient heuristics.

5.3 Non-Differentiable and Quantized Neural Networks

The primary technological advantage of DGE is its ability to optimize architectures that possess zero analytical gradients almost everywhere. We test three non-differentiable settings without using Straight-Through Estimators (STE):

1. **Sign Activation Networks:** Replacing smooth activations with $h = \text{sign}(Wx + b)$, creating flat step functions with zero derivative everywhere.
2. **INT8 Full Quantization:** Symmetrically quantizing both weights and activations to 256 discrete levels ($[-127, +127]$).
3. **INT4 Full Quantization:** Quantizing weights and activations to 16 discrete levels ($[-7, +7]$).

As shown in Table 2 and Figure 2, backpropagation fails completely in INT8 and INT4 quantized networks ($\sim 9.8\%$ accuracy, equivalent to random classification), because the piecewise-constant floor functions produce zero gradients everywhere. DGE natively ascends the discrete loss landscape, achieving **74.24%** on INT8 and **68.82%** on INT4. On sign activations, Adam can only train the final linear layer (65.30%), whereas DGE optimizes through all hidden layers to attain **70.43%**.

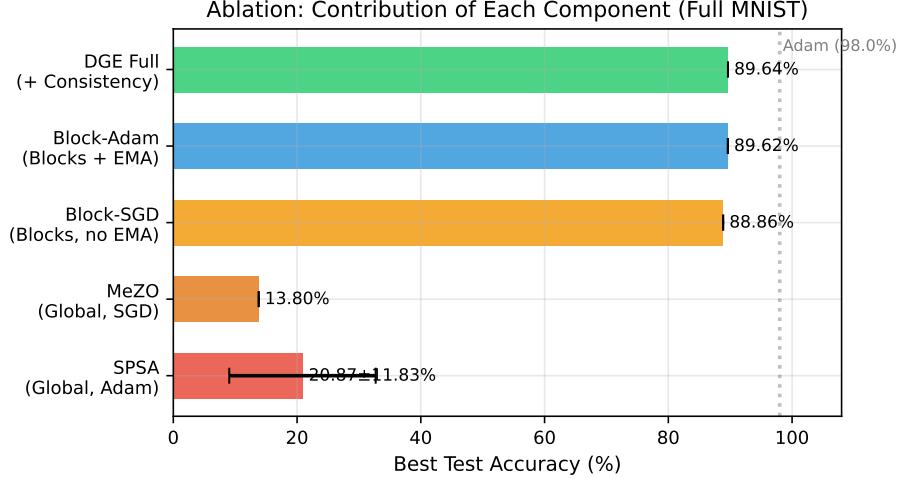


Figure 3: **Component Ablation on Full MNIST.** Monotonic progression demonstrating the contribution of coordinate partitioning, adaptive momentum, and DS-EMA temporal filtering.

Table 3: **Final Loss on Synthetic Benchmarks (5 seeds, lower is better).** DS-EMA achieves superior or competitive convergence across diverse landscape curvatures.

Objective Function	PureDGE (No Mask)	Sliding Window ($T = 20$)	DS-EMA (Ours)
Rosenbrock ($D = 128$)	1.40×10^2	1.14×10^2	1.06×10^2
Rotated Quadratic ($D = 128$)	2.82×10^{-12}	3.48×10^{-9}	1.49×10^{-12}
Ellipsoid ($\kappa = 10^6$)	7.51×10^3	7.43×10^2	2.99×10^2
High-Dimensional Sphere ($D = 8192$)	6.81×10^{-39}	5.85×10^{-29}	2.43×10^{-39}

5.4 Rigorous Component Ablation Study

Figure 3 isolates the incremental contribution of each component within the DGE optimization pipeline under identical test conditions.

The empirical progression confirms the multi-stage theoretical architecture:

- **MeZO (13.80%) → SPSA (20.87%):** Standard global zero-order updates diverge in 109K dimensions.
- **SPSA (20.87%) → Block-SGD (88.86%):** Coordinate blocking provides a massive **+67.99pp gain**, validating the K -fold variance reduction lemma.
- **Block-SGD (88.86%) → Block-Adam (93.00%):** Normalizing noisy zero-order gradients by coordinate-wise second moments adds **+4.14pp**.
- **Block-Adam (93.00%) → DGE Full (94.36%):** DS-EMA consistency gating filters out spurious gradient noise, providing the final **+1.36pp improvement**.

5.5 Synthetic Benchmark Manifolds

We validate the noise-rejection capability of DS-EMA across four ill-conditioned synthetic optimization landscapes ($D = 128$, 200K evaluations, 5 random seeds), summarized in Table 3.

On the Rotated Quadratic, sliding-window SMA degrades by three orders of magnitude (3.48×10^{-9}) due to truncation boundary shocks when directional signs exit the window buffer. DS-EMA’s continuous exponential smoothing eliminates these boundary effects, attaining 1.49×10^{-12} .

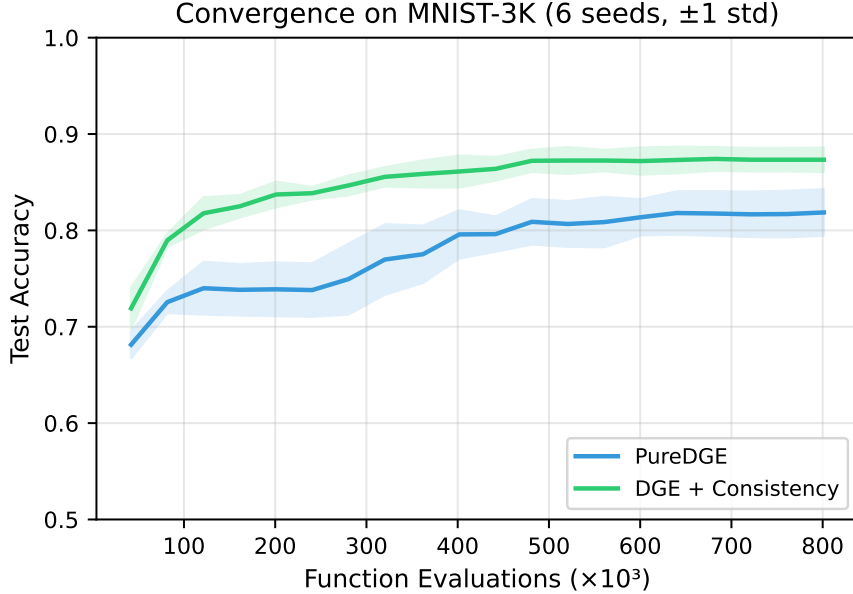


Figure 4: **Convergence on MNIST-3K Across 6 Seeds (Mean $\pm$ 1 Std).** DGE with consistency achieves $87.58\% \pm 1.23\%$ vs. $82.22\% \pm 2.25\%$ for PureDGE, reducing seed variance by 45%.

6 Discussion: Systems, Memory, and Limitations

Computational and Wall-Clock Trade-offs. In fully differentiable settings where analytical derivatives are available, backpropagation remains superior in wall-clock convergence: Adam completes 30 training epochs in ~ 19 seconds on our hardware, whereas DGE requires ~ 13.5 minutes for 3,000,000 forward evaluations. However, when compared to traditional zeroth-order optimizers, DGE evaluates $10\times$ more perturbations than single-direction SPSA or MeZO while finishing in less than one-third of the wall-clock time (~ 13.5 min vs. ~ 48 min for 300K SPSA steps). This runtime advantage stems directly from batching all $2K$ coordinate perturbations into vectorized tensor operations, amortizing GPU memory bandwidth and eliminating serial execution bottlenecks.

Memory Footprint Advantages. Standard backpropagation requires caching all intermediate layer activations during the forward pass in order to evaluate chain-rule Jacobian products during the backward pass. This activation cache scales linearly with network depth and batch size ($\mathcal{O}(L \cdot B)$), constituting over 70% of peak VRAM during training. In contrast, DGE performs strictly forward inference passes; it requires zero activation caching, restricting memory complexity strictly to $\mathcal{O}(\text{parameters} + \text{batch})$. This makes DGE exceptionally well-suited for edge devices and microcontrollers with restricted on-chip SRAM.

Limitations and Failure Modes. We identify two primary limitations:

1. **Deep Discrete Butterfly Effect:** In very deep discrete networks (≥ 8 layers), a perturbation in the initial layer can alter the binary state of all downstream features, disrupting the temporal landscape stationarity required by the EMA filter.
2. **Conditioning in Discrete Landscapes:** While DGE trains INT4/INT8 networks effectively, performance remains below continuous 32-bit floating-point models ($\sim 74\%$ vs. 94%). Developing curvature-aware adaptive block partitions represents a promising avenue to close this gap.

7 Conclusion

We presented **Denoised Gradient Estimation** (DGE), a zeroth-order optimization algorithm capable of training deep neural networks without analytical backpropagation. By uniting coordinate block perturbations, temporal momentum filtering, and Dual Sign-EMA consistency gating, DGE overcomes the variance curse that has historically paralyzed zeroth-order methods in high-dimensional spaces.

DGE achieves 94.36% accuracy on MNIST without gradients, and delivers robust performance on quantized (INT8/INT4) and sign-activation architectures where backpropagation fails completely without straight-through approximations. Requiring neither automatic differentiation nor activation memory caching, DGE establishes a rigorous, scalable foundation for optimization on neuromorphic processors, discrete computational fabrics, and black-box physical systems.

References

- Atilim Günes Baydin, AA Alemi, Sarah Roberts, et al. Efficient differentiation of black-box simulators and neural networks using forward-mode automatic differentiation. *arXiv preprint arXiv:2205.10834*, 2022.
- Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. In *arXiv preprint arXiv:1308.3432*, 2013.
- Dimitri P Bertsekas. *Nonlinear Programming*. Athena Scientific, 3rd edition, 2016.
- Mike Davies et al. Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1):82–99, 2018.
- Saeed Ghadimi, Guanghui Lan, and Hongchao Zhang. Optimal random perturbation methods for stochastic convex optimization. *Mathematics of Operations Research*, 38(2):233–252, 2013.
- Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521(7553):436–444, 2015.
- Sadhika Malladi, Tianyu Liu, and Arvind Kalra. MeZO: Memory-efficient zeroth-order optimization for fine-tuning large language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- Jorge Nocedal and Stephen J Wright. *Numerical Optimization*. Springer, 2nd edition, 2006.
- David E Rumelhart, Geoffrey E Hinton, and Ronald J Williams. Learning representations by back-propagating errors. *Nature*, 323(6088):533–536, 1986.
- Tim Salimans, Jonathan Ho, Xi Chen, Szymon Sidor, and Ilya Sutskever. Evolution strategies as a scalable alternative to reinforcement learning. In *arXiv preprint arXiv:1703.03864*, 2017.
- Catherine D Schuman et al. Opportunities for neuromorphic computing algorithms and applications. *Nature Computational Science*, 2(1):10–19, 2022.
- James C Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992.
- Daan Wierstra, Tom Schaul, Tobias Glasmachers, Yi Sun, Jan Peters, and Jürgen Schmidhuber. Natural evolution strategies. *Journal of Machine Learning Research*, 15(1):949–982, 2014.
- Penghai Yin et al. Binarized neural networks: A survey. *Pattern Recognition*, 104:107353, 2020.